

EXPERIMENT 1A

SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

Aim :

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications.

Algorithm :

1. Start the program.
2. Import the required libraries (if needed).
3. Declare and initialize the required variables or data.
4. Perform the required operations or processing.
5. Display the output/results.
6. Stop the program.

Code :

```
#To print  
print("Welcome to Data Analytics Lab")
```

Welcome to Data Analytics Lab

#declare variable and perform mathematical functions

a = 20

b = 10

print(a+b)

print(a-b)

print(a*b)

print(a/b)

30

10

200

2.0

: *#declare variables and print it*

name = "John"

dept = "AI&DS"

cgpa = 8.9

print("Name:", name)

print("Department:", dept)

print("CGPA:", cgpa)

Name: John

Department: AI&DS

CGPA: 8.9

```
#import numpy and perform mean and sum operations  
import numpy as np  
a = np.array([10,20,30,40,50])  
print(a)  
print(np.mean(a))  
print(np.sum(a))
```

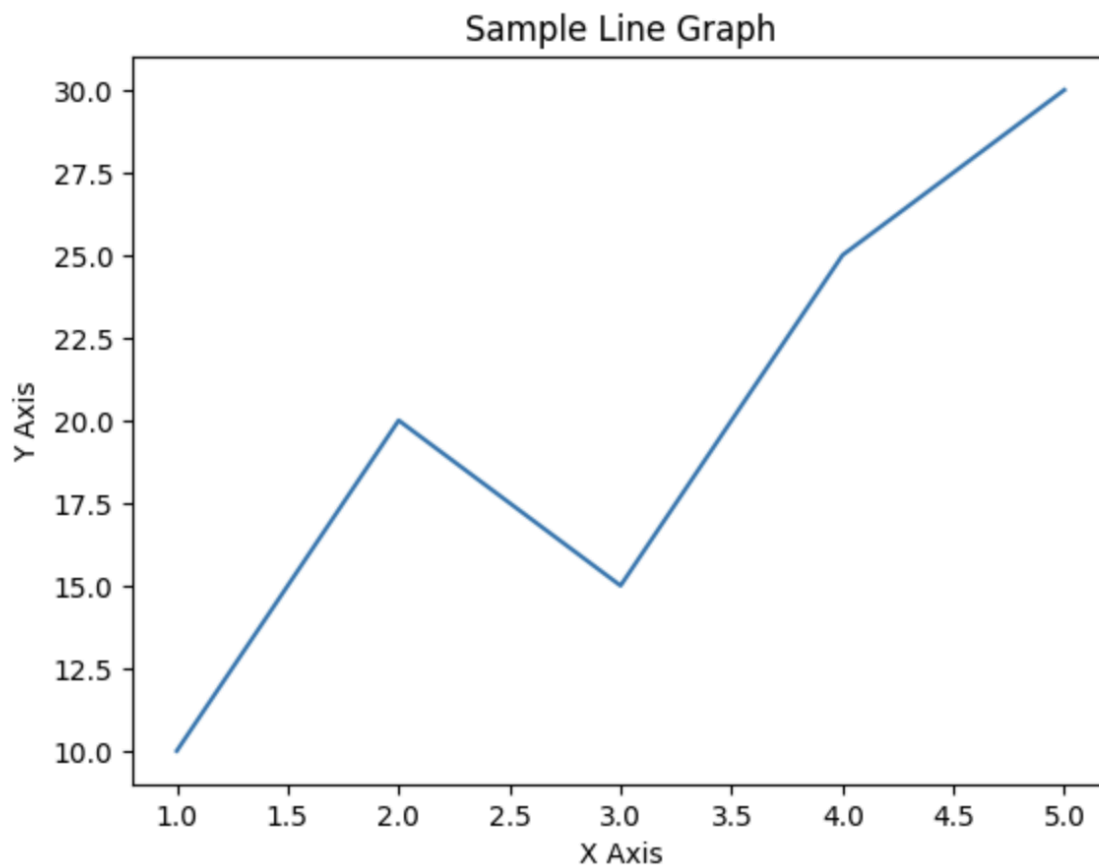
```
[10 20 30 40 50]  
30.0  
150
```

```
#import pandas as pd and create a dcitionary  
#and make it as a DataFrame  
import pandas as pd  
data = {  
    'Name': ['John', 'David', 'Alex'],  
    'Marks': [85, 90, 95]  
}  
df = pd.DataFrame(data)  
print(df)
```

	Name	Marks
0	John	85
1	David	90
2	Alex	95

```
: #To plot a simple line graph
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]

plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```



Result :

The program to perform basic mathematical operations is completed successfully